# Harry Brook Runs Prediction

AI Course Project Proposal | FUUAST, Karachi

## 1. Selected Player

Harry Brook (English cricketer, right-handed batter)

## 2. Objective

Predict how many runs Harry Brook will score in his next One Day International (ODI) innings.

## 3. Approach

Supervised learning (the output is a numerical value: runs scored). This is a regression problem.

## 4. Model

Decision Tree Regressor (main model), compared with Random Forest Regressor.

## 5. Reason

- Supervised learning: each previous ODI innings has a known runs-scored value, so the data is labelled.
- Decision Tree: can use factors such as opponent, venue, batting position and recent form; it can model non-linear patterns and is relatively easy to interpret.
- Random Forest: combines multiple decision trees, which can reduce overfitting and provide more stable predictions.

## 6. Dataset (Authentic Sources)

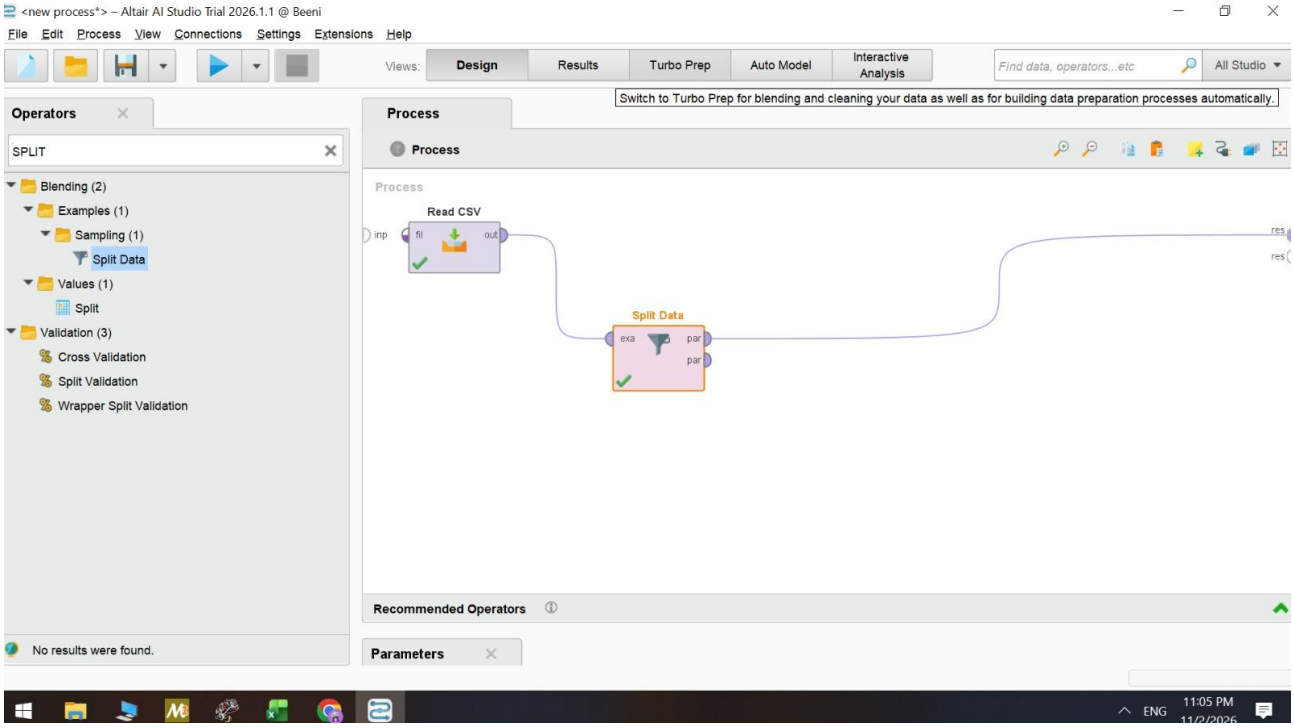
**Main source (ICC):** Harry Brook player profile and match-by-match records
https://www.icc-cricket.com/rankings/66374/harry-brook

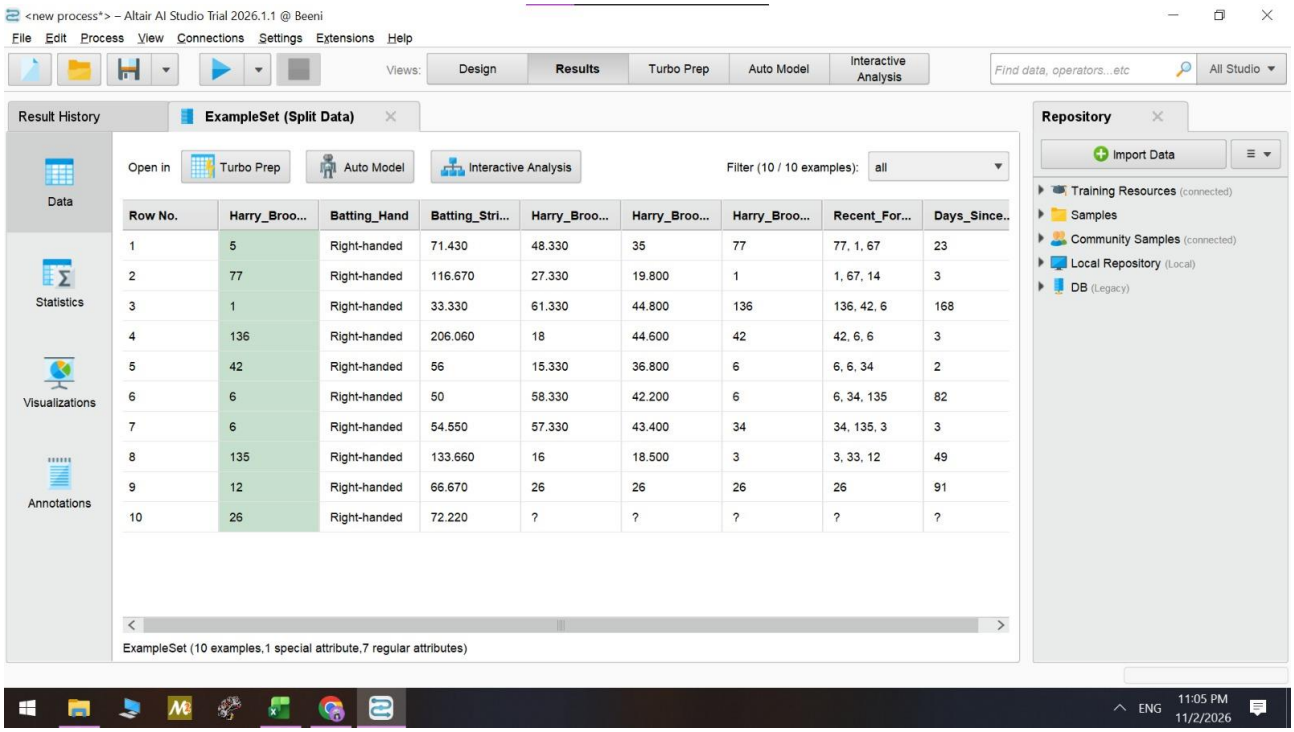
**Verification source (ESPNcricinfo Statsguru):** International cricket batting statistics and match records
https://stats.cricinfo.com/cricinfo/engine/stats/index.html

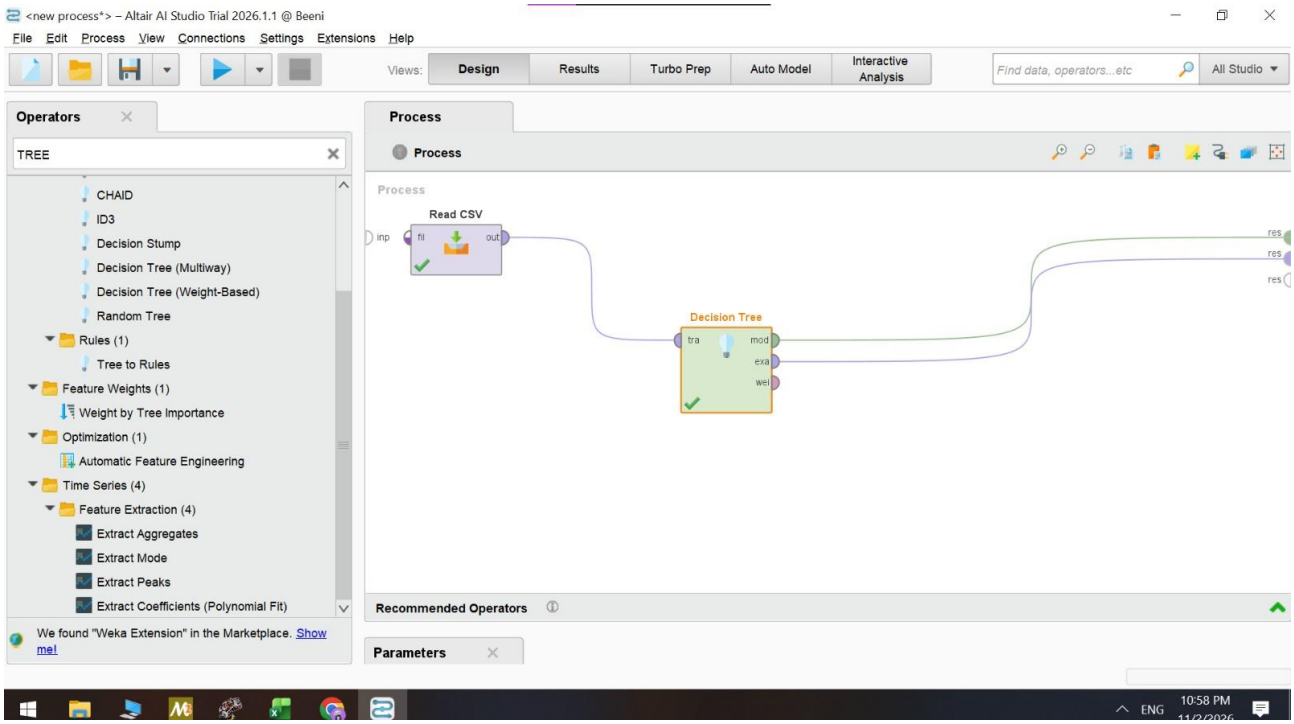
**Additional official source (ECB):** Player profile and playing role
https://www.ecb.co.uk/players/3375

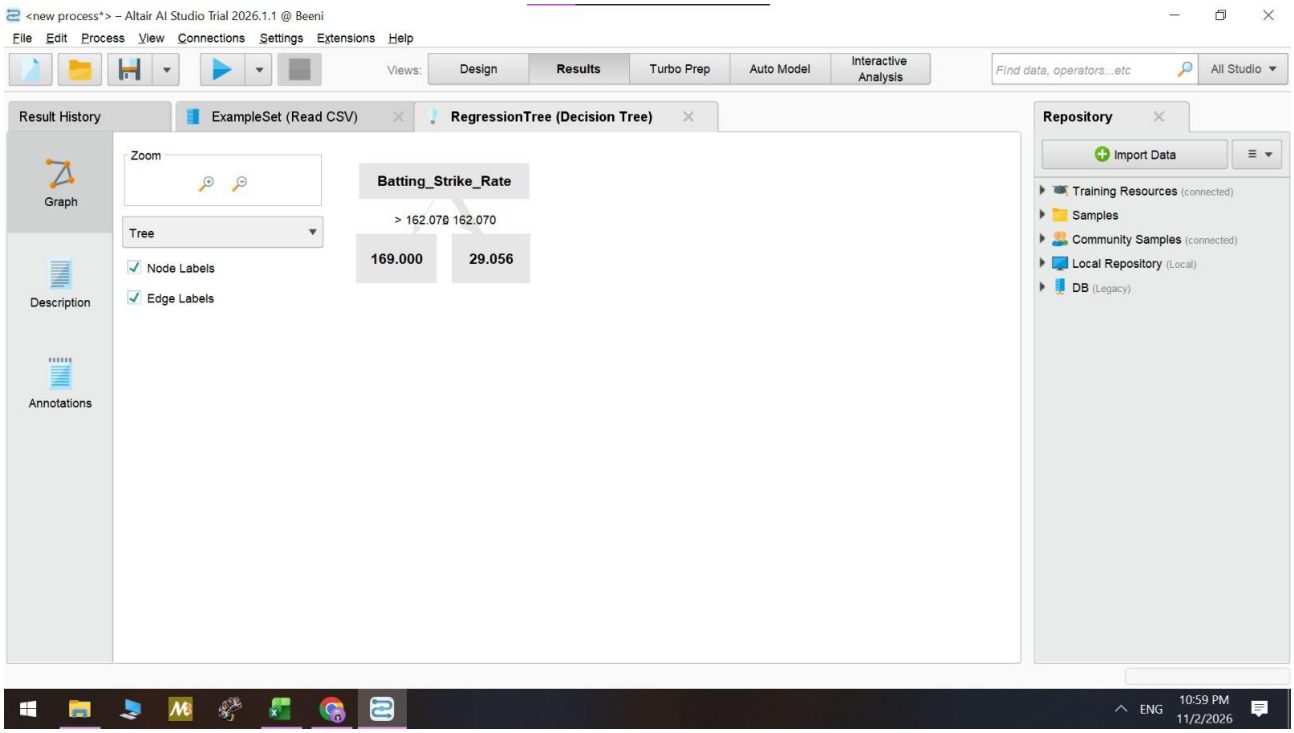
ODI innings records will be collected with features such as opponent, venue, batting position, innings number and recent performance. The target column will be runs scored in each innings. Records will be checked against the ICC and ESPNcricinfo sources before model training.

# RESULT SCREENSHOTS




<new process*> – Altair AI Studio Trial 2026.1.1 @ Beeni
File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
Find data, operators...etc
All Studio
Operators
TREE
CHAID
ID3
Decision Stump
Decision Tree (Multiway)
Decision Tree (Weight-Based)
Random Tree
Rules (1)
Tree to Rules
Feature Weights (1)
Weight by Tree Importance
Optimization (1)
Automatic Feature Engineering
Time Series (4)
Feature Extraction (4)
Extract Aggregates
Extract Mode
Extract Peaks
Extract Coefficients (Polynomial Fit)
We found "Weka Extension" in the Marketplace. Show me!
Process
Process
Process
Read CSV
inp
fil
out
Decision Tree
tra
mod
exa
wei
res
res
res
Recommended Operators
Parameters
ENG
10:58 PM


<new process*> – Altair AI Studio Trial 2026.1.1 @ Beeni
File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
Find data, operators...etc
All Studio
Result History
ExampleSet (Read CSV)
RegressionTree (Decision Tree)
Graph
Description
Annotations
Zoom
Tree
Node Labels
Edge Labels
Batting_Strike_Rate
> 162.070 162.070
169.000
29.056
Repository
Import Data
Training Resources (connected)
Samples
Community Samples (connected)
Local Repository (Local)
DB (Legacy)
ENG
10:59 PM
11/2/2026